PhenoCropNet: A Phenology-Aware-Based SAR Crop Mapping Network for Cloudy and Rainy Areas

Lei Lei, Xinyu Wang, *Member, IEEE*, Xin Hu, *Member, IEEE*, Liangpei Zhang, *Fellow, IEEE*, and Yanfei Zhong, *Senior Member, IEEE*

***Abstract*— Crop mapping in a cloudy area is always a challenge due to the lack of time-series clear optical satellite imagery. Making use of time-series synthetic aperture radar (SAR) imagery that is immune to cloud contamination is essential and promising for seamless and large-area crop mapping. However, existing deep learning (DL)-based crop classification methods give the extracted phenological features equal weights, without considering the different contributions of phenological features of the different crop growth periods. In this article, a phenology-based crop mapping network (PhenoCropNet) is proposed to extract the discriminative features from the two levels, including the key phenological dates in the phenological periods and key phenological periods in the whole growth stages. PhenoCropNet includes a phenological calendar information injection (PAI) module that divides the satellite imagery time series (SITS) into multiple sequences according to the phenological calendar information, and a hierarchical attention network structure that uses the two-level bidirectional gated recurrent unit-based self-attention (BiGRUA) modules to automatically extract the features containing the most important phenological information of key phenological dates and key phenological periods. The proposed PhenoCropNet was verified in Hubei province in China, around 185 933 km^2, a typical cloudy area in China, for rapid winter crop mapping based on temporal Sentinel-1 SAR imagery. The mapping result shows that the $F1$-score of PhenoCropNet for winter crop mapping could achieve 0.90, showing great potential in large-scale and seamless crop mapping. The code is available on request: https://github.com/LL0912/PhenoCropNet.**

***Index Terms*— Crop mapping, deep learning (DL) method, phenological information, time series Sentinel-1.**

I. Introduction

DESCRIBING the spatial distribution of crop types via crop mapping [1], [2], [3], [4] is an important basis for agricultural geoinformation, paving the way for subsequent practical application and scientific analysis, such as crop management [5], food security supervision [6], environmental protection [7], and biodiversity research [8]. While it remains to be a great challenge for crop mapping in cloudy area, such as South China, where it is cloudy and rainy for months on end during the crop growth stages, making it unavailable to acquire clear optical satellite imagery time series (SITS) to capture the phenological feature of crop mapping [9]. Synthetic aperture radar (SAR), that can provide all-day and all-weather observation by active remote sensing, is the potential for seamless crop type mapping [10], [11], [12], [13] and crop intensity mapping [14] at large scale regardless of the local climate condition.

An important challenge when it comes to crop mapping using SAR SITS is the similarity of phenological evolution of different crop types, causing confusion between crops [15]. The phenological evolution in SAR crop mapping refers to the time-series backscattering of different polarization modes in the SAR system, such as VV or VH. The critical point of crop mapping using SAR SITS is the extraction of discernible features from the phenological evolution. Traditional methods extract the features by manually-designed various phenological metrics, such as the index designed for a particular crop type based on SAR phenological evolution [16], the slope or the variance of the phenological evolution [17], the starting and ending date of the crop growing season, or the length of the growth periods [15]. Machine learning (ML) methods are also adopted to select the most representative features from a large amount of temporal based on some optimal feature selection index, such as the out-of-bag score or Gini score in the random forest [19], [20], [21], [22]. The advantage of ML methods is the interpretability of the model by ranging the important features, which has been applied in crop classification [22] and relative phenological parameter inversion, such as crop yield estimation [22]. By adjusting the input features of the ML method, the important SAR-specific features can be determined, for example, the optimal asymmetric coherence of SAR and polarimetric SAR is found better than temporal coherence [23]. The crop classification accuracy of both categories above depends on the manually designed features, which might be out of operation when there are no conspicuous phenological characters.

To mitigate the limitation of manually crafted features, recently, deep learning (DL) techniques for their strong

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Lei Lei, Liangpei Zhang, and Yanfei Zhong are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, and Hubei Provincial Engineering Research Center of Natural Resources Remote Sensing Monitoring, Wuhan University, Wuhan 430079, China (e-mail: leilei912@whu.edu.cn; zlp62@whu.edu.cn; zhongyanfei@whu.edu.cn).

Xinyu Wang is with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China (e-mail: wangxinyu@whu.edu.cn).

Xin Hu is with Guangzhou Urban Planning and Design Survey Research Institute, Guangzhou 510060, China.

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feature extraction capability have been employed for crop mapping using time-series SAR imagery [24], [25], [26], [27], [28], [29], [30]. To process time-series information, 1-D convolutional neural networks (1D-CNNs) [22] and the series of recurrent neural networks (RNNs), including long short-term memory (LSTM) models [27] and gated recurrent units (GRUs) [28], have been widely adopted to realize pixel-wise classification, due to their ability to build the long-term relationships of time-series data. Another mainstream of DL method is the multilevel attention mechanism, which has been adopted to excavate multiscale spatial-temporal features of time-series SAR for crop classification [29]. While these methods do not reveal the principle of DL methods, and fail to answer the question that which dates or period is more important for crop classification. Furthermore, in order to build trust in DL methods, some recent research has engaged in the interpretability of the DL model [31], [32]. For example, the self-attention mechanism is combined with LSTM to capture the extracted features of more importance [31]. And grad-weighted class activation mapping (Grad-CAM) [33] has been adapted to time-series remote sensing imagery for unsupervised classification, therefore, confused crop types could be explained by analyzing the important date to the model [32].

Although DL methods have achieved excellent performance in crop mapping, these methods, however, do not consider the phenological knowledge of crop growth during the model designing process or the model training process. Phenological knowledge refers to the prior knowledge about crop growth, such as crop calendar of different growth period, and the important growth period to differentiate crop from other land cover types [34], [35]. For example, previous studies have demonstrated the importance of different crop growth period to crop classification [36], [37], [38], and crop biophysical parameter assessment [39]. Specifically, the flowering period is found crucial for rapeseed extraction from optical remote sensing images, during which the hue of rapeseed parcel shows a yellow–green color [40], while the early stage is found more sensitive for crop biophysical parameter monitoring using temporal Sentinel-1 backscattering parameters [37]. Adding existing phenological knowledge to the DL model will not only be beneficial to improve the ability of the model to distinguish different crop types, but also helpful to escalate the capability of the model to perceive the important phenological periods, making classification and interpretation feasible concurrently. Therefore, how to add the phenological knowledge is the main challenge encountered by existing DL models.

The objective of this study is to add phenological knowledge to the DL model to discriminate crop with similar phenological evolution and reveal phenology-related mechanisms perceived by the model for better classification and interpretation. PhenoCropNet is proposed in this study with the following three improved aspects to fuse phenological knowledge.

1) Injecting crop calendar information. Since SAR could provide images covering the whole phenological calendar circle, making full use of the phenological information of the crop growth calendar is favorable to differentiate crop with similar phenological characters and facilitate the following feature extraction of key phenological periods. Specifically, to add phenological calendar information, the input complete time-series SAR backscattering is divided into several shortest sequences without intersection to represent different crop growth stages.
2) Designing a hierarchical attention network. A hierarchical attention network structure is superior to extracting features from both the key dates and key periods at a local and global perspective. Since the attention weights of the hierarchical structure that show the importance score of phenological dates and periods can be learned automatically, the phenological knowledge of important growth dates and periods could be added. In detail, the first and the second layers are to excavate the important features of key dates and key periods, respectively.
3) Designing an efficient basic module. To effectively model the long-term relationship of SAR SITS, a bi-directional GRU is employed as the basic unit for comparable performance but fewer computing resources than LSTM. Moreover, the self-attention mechanism is employed to concentrate and refine the most important deep features.

Given the great data procession pressure from the GB range data of SAR SITS, a rapid workflow of large-scale crop mapping online based on the Google Earth Engine (GEE) platform is proposed and implemented in Hubei province, covering around 185 933 km^2 acreage, for winter crop mapping combined with the proposed method. The results confirm the effectiveness of each module in PhenoCropNet and the superiority of PhenoCropNet in both early- and in-season crop mapping. Moreover, the visualization analysis of the PhenoCropNet results verifies the coincidence of the extracted phenological weights with the phenological prior knowledge, helping the understanding of the network. To summarize, our contributions are as follows.

1) A phenology-aware-based crop mapping network, named PhenoCropNet, including a phenological calendar information injection (PAI) module and a hierarchical phenology self-attention network structure is proposed to extract the discriminate features from two levels: key dates and key growth periods to distinguish crops with similar phenology.
2) The PAI module is designed to endow SITS with the phenological meaning of crop growth periods to facilitate the network to extract the discriminative deep representation of crop growth periods.
3) Hierarchical phenology self-attention network structure taking bi-directional GRU and self-attention mechanism as the basic unit is designed to help the model capture the key date and key growth periods in a local and global perspective for crop classification.

II. Methodology

Formally, the aim is to assign a label for each pixel $x_i \in \mathbb{R}^{t\times m}$ belonging to the image $X \in \mathbb{R}^{H\times W\times t\times m}$ to acquire the crop distribution map $Y \in \mathbb{R}^{H\times W\times C}$, where H, W, C, t, and m are the height and the width of the imagery,

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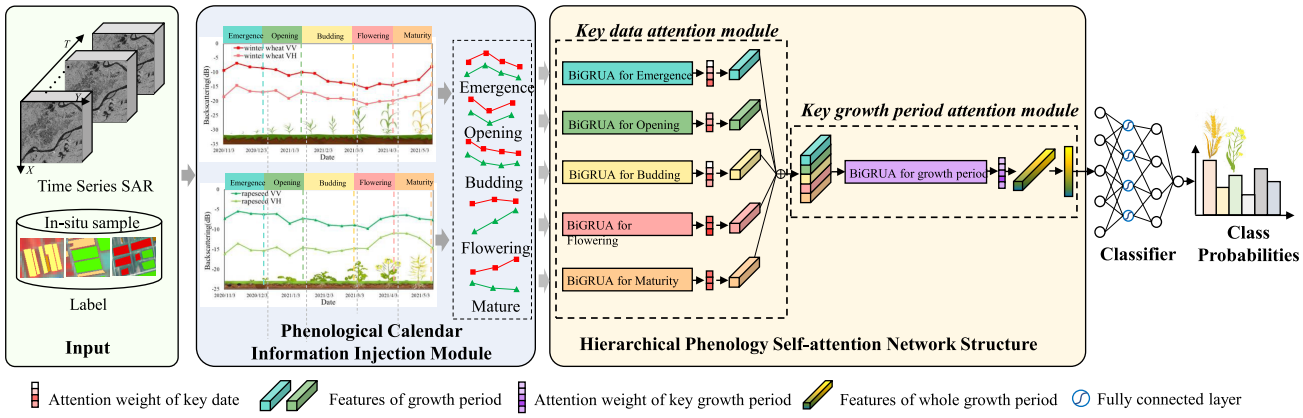


Fig. 1. Framework of the phenology-aware-based crop mapping network.

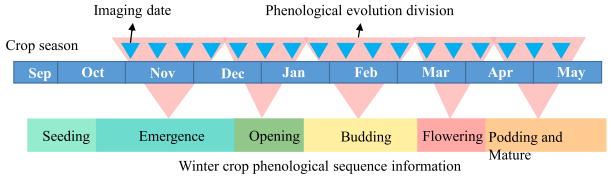


Fig. 2. Illustration of the phenological sequence information injection module.

the number of class categories, time-series length, and the number of input band, respectively. SAR can acquire complete time-series imagery in cloudy area, but the contribution of the features at each date to crop classification is different. Paying adequate attention to and excavating the important features is conducive to improving the discriminability of the classifier. PhenoCropNet is designed to extract the important features from two levels: the key dates and key growth periods, with the assistance of phenological calendar information, as Fig. 1 shows.

A. PAI Module

Phenological calendar information representing the different crop growth stages is crucial for crop mapping because the discriminative features of the crop show a dynamic change with the development of the crop. In the different phenological periods, the contribution of the features at different dates also varies. The PAI module is proposed to divide the SITS according to the phenological calendars of the crops. With the access of the imaging dates and the local phenological calendar, time-series SAR can be clipped into subsequence, such as the emergence, opening, budding, flowering, and podding and maturity for rapeseed as shown in Fig. 2. If there are more than one crop type and the crop growth periods are overlapped for various crop types, the temporal SITS are cropped into the sequence with comprehensive phenological calendar information. The formula of PAI can be expressed as follows, where $\boldsymbol{s}_i$ is the clipped subsequence of time-series SAR, $\boldsymbol{c}_j$ represents a group of day of year (DoY) of crop growth period for the crop type j, $l(\cdot)$ represents the length of crop growth period. The equations illustrate the basic idea of PAI, which splits the complete time-series SAR into the shortest sequence without overlap to represent the crop growth period of different types. Through this module, the prior knowledge about the time schedule of crop growth can be injected into the model, to facilitate the network to make full use of the phenological information automatically

$$s_i = \begin{cases} \text{if } \boldsymbol{c}_1 \cap \boldsymbol{c}_2 \cap, \ldots, \cap \boldsymbol{c}_j = \emptyset: \\ \boldsymbol{c}_1 \text{ or } \boldsymbol{c}_2 \text{ or } \boldsymbol{c}_j \\ \text{if } \boldsymbol{c}_1 \cap \boldsymbol{c}_2 \cap, \ldots, \cap \boldsymbol{c}_j \neq \emptyset: \\ \min(l(\boldsymbol{c}_1), \ldots, l(\boldsymbol{c}_j)) \text{ or } l(\boldsymbol{c}_1 \cap \boldsymbol{c}_2 \cap, \ldots, \cap \boldsymbol{c}_j) \\ -\min(l(\boldsymbol{c}_1), \ldots, l(\boldsymbol{c}_j)) \end{cases} \tag{1}$$

$$t = \sum_i l(s_i). \tag{2}$$

B. Hierarchical Phenology Self-Attention Network Structure

PhenCropNet has a two-level hierarchical attention network structure to extract the discriminate features of the key dates and key periods. Each level in the hierarchical network structure is composed of several bidirectional GRU-based self-attention (BiGRUA) modules, the details of which are to be explained in Section II-C. In Section II-B, the functions and details of each layer in the hierarchical attention network structure are to be stated.

1) Key Date Attention Level: The first level takes the subsequence corresponding to different crop growth stages as the input. Each subsequence is utilized by an independent BiGRUA unit to extract the important features of the key date in the growth period. The motivation to design this module is based on the inspiration that although SAR can monitor the crop growth status in a continuous way, the separability of crop varies for different dates. Locating at the key date with the most obvious features can force the model to concentrate on the date with high separability and excavate the features at that date. In detail, if the SAR SITS is clipped into subsequence $s_i, i \in [1, q]$, the time step of each subsequence could contain different numbers of time steps, and the key date attention level would be composed of q BiGRUA module. Each sequence s_i is input into the corresponding independent BiGRUA module to obtain the key date-focusing features $\boldsymbol{O}_i^d \in \mathbb{R}^{N \times 1 \times v}, i \in [1, q]$, where N and v are the batch size of training samples and the number of hidden neurons in the first-level BiGRUA module.

2) Key Growth Period Attention Module: The second level of the key growth period consists of one single BiGRUA module, taking the output of each phenological period concatenated

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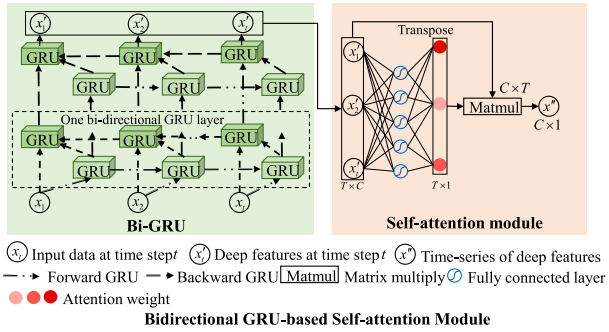


Fig. 3. Structure of BiGRUA module.

as the input $\boldsymbol{O}^d \in \mathbb{R}^{N\times q\times v}$, to obtain the key phenological period features $\boldsymbol{O}^g \in \mathbb{R}^{N\times 1\times V}$, where V is the number of hidden neurons in the second-level BiGRUA module. The motivation for designing the second level is based on the phenological knowledge that the appearance of a crop varies at different growth periods, resulting in different separability on remote sensing imagery. Previous studies have demonstrated the importance of key growth periods in crop classification. In the second level, by fusing the features of different growth stages, the key growth stage would be located from a global perspective.

C. Bidirectional GRU-Based Self-Attention Module

BiGRUA, as Fig. 3 shows, is the specific module designed to extract features from the key date and key growth periods of SITS, which consists of two structures, multiple bidirectional GRU layers, and a self-attention layer. The advantage of a bidirectional GRU-based unit is that it is an ideal structure to process time-series date, avoiding forgetting effectively and grasping context information of temporal data. The merits of the self-attention mechanism are the automation and the convenience of obtaining the key date and key growth period.

1) Bidirectional GRU-Based Unit: Each GRU layer uses one GRU unit iteratively to decide the information to be preserved or abandoned in the SAR SITS automatically via the gate mechanism, having the ability to excavate the abstract features of temporal SAR data. One GRU unit contains one reset gate, one update gate, and one renewed hidden cell, as shown in the following equations:

$$\boldsymbol{gate}_{\text{reset}}^{\text{forward}} = \sigma\left(\boldsymbol{w}_{rx}\boldsymbol{x}_t^{l-1} + \boldsymbol{w}_{rh}\boldsymbol{h}_{t-1}^{l} + \boldsymbol{b}_r\right) \tag{3}$$

$$\boldsymbol{gate}_{\text{update}}^{\text{forward}} = \sigma\left(\boldsymbol{w}_{ux}\boldsymbol{x}_t^{l-1} + \boldsymbol{w}_{uh}\boldsymbol{h}_{t-1}^{l} + \boldsymbol{b}_u\right) \tag{4}$$

$$\boldsymbol{n}_t = \tanh\left(\boldsymbol{gate}_{\text{reset}}^{\text{forward}} \cdot \left(\boldsymbol{w}_{nh}\boldsymbol{h}_{t-1}^{l}\right) + \boldsymbol{w}_{nx}\boldsymbol{x}_t^{l-1} + \boldsymbol{b}_n\right) \tag{5}$$

$$\boldsymbol{h}_t^{\text{forward}} = \boldsymbol{h}_t^{l} = \left(1 - \boldsymbol{gate}_{\text{update}}^{\text{forward}}\right) \cdot \boldsymbol{n}_t + \boldsymbol{gate}_{\text{update}}^{\text{forward}} \cdot \boldsymbol{h}_{t-1}^{l} \tag{6}$$

where $\boldsymbol{w}_{rx}$, $\boldsymbol{w}_{rh}$, $\boldsymbol{w}_{ux}$, $\boldsymbol{w}_{uh}$, $\boldsymbol{w}_{nh}$, $\boldsymbol{w}_{nx}$, $\boldsymbol{b}_r$, $\boldsymbol{b}_u$, and $\boldsymbol{b}_n$ are the learnable weights, σ and tanh are the activation function. Both the reset gate $\boldsymbol{gate}_{\text{reset}}^{\text{forward}}$ and the update gate $\boldsymbol{gate}_{\text{update}}^{\text{forward}}$ are derived from the input $\boldsymbol{x}_t^{l-1}$ at the current t time step of the last $l-1$ layer and the hidden state $\boldsymbol{h}_{t-1}^{l}$ at the previous $t-1$ time step of the current l layer. The reset gate determines how much of the previous hidden state to retain for the renewed hidden state, along with the input of the current time step. The update gate selects which parts of the previous hidden state to preserve for the current output while discarding the remainder. As for the missing parts, the inverse of the update gate is used to patch up the missing parts from the renewed hidden state. The hidden state $\boldsymbol{h}_t^{l}$ is the output of the current l layer at t time step and also the hidden state for the next $t+1$ time step. Through the gated recurrent structure, GRU can extract the features of time-series satellite imagery automatically.

To better build the interactive relationship in the long-term context, a bidirectional GRU is applied, which uses the output at the final time step as the first hidden state to build a reversed GRU layer. The output of the forward and backward GRU layer is concatenated as $[\boldsymbol{h}_t^{\text{forward}}, \boldsymbol{h}_t^{\text{backward}}]$, named $\boldsymbol{h}_t$ for the subsequent self-attention module, which is designed to better aggregate the effective features and disregard the useless features for classification.

2) Self-Attention Mechanism: The attention mechanism simulates the human vision system, which selectively chooses the conscious parts to process, leaving the trivial parts unprocessed when confronted with a predetermined purpose. The attention network implements the focus on a certain important part by adopting element-wise multiplication (Hadamard product) with the self-adaptive weight calculated previously. The ways to calculate the weight are various, such as a general function [42] and a nonlocal function [43]. In the proposed approach, the popularly employed general function is used to calculate the weights from the output of the bidirectional GRU layer. It comprises one fully connected layer followed by a softmax activation function to generate the attention weight self-adaptively. The attention weight is then multiplied with the bidirectional GRU output on the element-wise production to generate the final abstract feature. The mathematical formulas for this process are shown in the following equations:

$$\boldsymbol{a}_t = \text{softmax}(\boldsymbol{w}_{at}\boldsymbol{h}_t + \boldsymbol{b}_{at}) \tag{7}$$

$$\boldsymbol{h}_t^{*} = \boldsymbol{a}_t \cdot \boldsymbol{h}_t \tag{8}$$

$$\text{softmax}(x) = \frac{e^x}{\sum e^x} \tag{9}$$

where $\boldsymbol{w}_{at}$ and $\boldsymbol{b}_{at}$ are the trainable parameter vectors.

D. Crop Mapping Classifier

A single-layer perception is employed as the classifier to process the output of the hierarchical attention network by assigning a label to each pixel, as follows:

$$\begin{aligned} \boldsymbol{O}^h &= \text{ReLU}(\boldsymbol{w}_{ht}\boldsymbol{O}^g + \boldsymbol{b}_{ht}) \\ \boldsymbol{p} &= \boldsymbol{w}_p\boldsymbol{O}^h + \boldsymbol{b}_h \end{aligned} \tag{10}$$

where $\boldsymbol{w}_{ht}$ and $\boldsymbol{b}_{ht}$ are the learnable parameters, $\boldsymbol{O}^h \in \mathbb{R}^{N\times v'}$ is the output of the single-layer perception with the feature dimension of v', $\boldsymbol{p} \in \mathbb{R}^{N\times C}$ is the predicted probability of the model, and C is the number of class categories.

The cross-entropy loss function is the optimization objective to train the weight of PhenoCropNet via the forward and backward propagation algorithm. L2-norm regularization is

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also added to prevent overfitting. The overall loss function follows:

$$L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} p_i^c \log(l_i^c) + \lambda \|\boldsymbol{w}\|^2 \tag{11}$$

where $\boldsymbol{w}$ represents the parameters in PhenoCropNet, and λ is the L2 regularization parameter.

E. Rapid Large-Scale Online GEE Crop Mapping Using Temporal Sentinel-1 Imagery

Large-scale crop mapping supported by the time-series remote sensing imagery poses a challenge to the data storage and computing resources, requiring the GB range storage system and a high-performance processor. This problem can be dealt with by utilizing a cloud computing platform that allows large-scale geospatial analysis. The GEE platform was adopted in this study as it allows convenient acquisition and quick preprocessing of temporal Sentinel-1 imagery. However, the application programming interfaces (APIs) for classification are limited on the GEE platform, which does not support customized methods, such as DL networks. To solve the problem and realize large-scale online crop mapping quickly and intelligently, Google Drive and Google Colab were also utilized to achieve the data transmit workflow of time-series Sentinel-1 imagery preprocession and analysis. The overall framework of crop mapping at a large scale, taking winter crop extraction in Hubei province as an example, is shown in Fig. 4. Generally speaking, this is an effective scheme that is not only appropriate for Hubei province, but also for other regions or countries, regardless of the weather conditions, to obtain crop mapping results both quickly and accurately.

III. Experiments and Analysis

A. Study Area and Dataset

The study area, Hubei province in China as shown in Fig. 5(a), spans 185 933 km^2 and lies in the middle reaches of the Yangtze River, renowned as one of the vital "bread baskets" of China. The acreage of winter wheat in Hubei province ranks twelfth and that of winter rapeseed ranks third among all the provinces in China. From the perspective of crop mapping in China, Hubei province is a typical and challenging area, due to the cloudy and rainy weather, diverse terrain conditions, and smallholder planting patterns. The west of Hubei province is mostly mountains, while the middle and eastern areas are plains. The long-lasting cloudy and rainy weather brings challenges to crop mapping using optical SITS data. The smallholder planting patterns also demand high-resolution satellite imagery. All these factors render Hubei province an exemplary and pivotal region for crop mapping in the middle and lower Yangtze River basin.

The Sentinel-1 satellite constellation consists of two identical radar imagery satellites in the same orbit, and can cover Hubei province in 12 days, providing adequate SITS data to monitor crop phenology. 17 time-series Sentinel-1A/B interferometric wide (IW) swath images of dual polarization (VV and VH) from October 22, 2020 to June 7, 2021 were collected from the ground range detected (GRD) dataset on the GEE platform [41]. The imaging dates corresponding to the crop growth periods are shown in Fig. 5(d).

TABLE I
Refined In Situ Sample Information in 2021 in the Study Area

Usage	Class Categories					
	WW	WR	OV	WT	LA	OT
Pixel						
Train	53800	7300	26178	67037	35491	140559
Valid	15367	2014	4939	27236	11669	27403
Test	37387	6280	19529	67444	24538	101784
Parcel						
Train	785	296	276	259	262	182
Valid	199	81	58	62	74	41
Test	590	214	227	197	194	123

Note: WW-winter wheat, WR-winter rapeseed, OV-other vegetable, WT-water, LA-leisure land, OT-others.

Sentinel-1 GRD dataset underwent preprocessing from the single look complex data (SLC) to the calibrated product on the GEE, including thermal noise removal, radiometric calibration, terrain rectification, and resampling to 10-m resolution. Other customized preprocessing steps, such as speckle filtering and radiometric terrain normalization were coded and implemented on the GEE platform considering the total data reaching 600 GB.

The in situ samples were collected during 24–26 April 2021, when the winter rapeseed was in the podding period and the winter wheat was in the earing period. The phenological calendar for the main winter crop types is shown in Fig. 5(d). The sampling route covered 11 counties in Hubei province, and 959 km in total, as shown in Fig. 5(a). For the label of crop type, it is difficult to interpret the crop type merely from remote sensing imagery, therefore, the label of all samples of crop types is from in situ windshield survey, during which the crop type of each parcel along the road is recorded on the mobile application named GPS Toolbox, with the location and photography, as shown in Fig. 5(b) and (c) following the planned route. Other categories such as water, fallow land, and artificial buildings were labeled by visual interpretation of high-resolution imagery. Due to the discrepancy of planting areas of different crop types, the number of crop samples is nearly impossible to maintain a balance. Considering the unequal distribution of different land-cover types, we remain all the labeled crop-type samples in the experiment. We kept the number of labeled parcels of each noncrop land-cover type that is annotated by interpretation of remote sensing imagery to be balanced, as shown in Table I. While due to the different shapes and sizes of different noncrop land-cover types, the number of labeled pixels might vary a lot. All the sampling points were checked twice with the help of Sentinel-2 imagery, labeled as independent parcels in ArcGIS 10.2, and saved in shapefile format. In the experiments, the parcels were divided into independent training, validation, and test datasets, to guarantee geographic separation. Table I enumerates the sample details by pixel and parcel counts.

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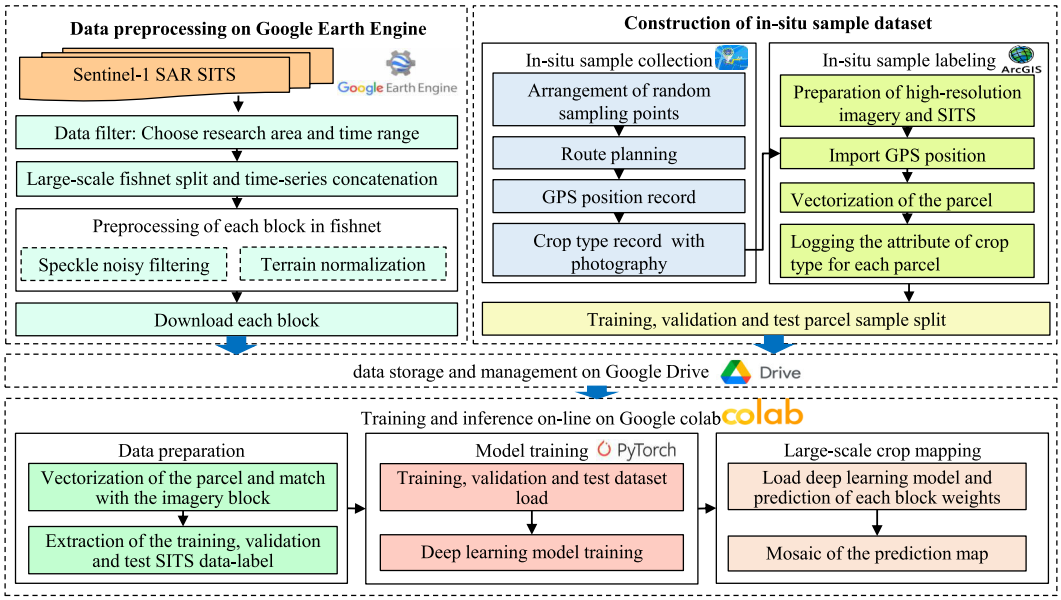


Fig. 4. Workflow of rapid large-scale online GEE crop mapping using temporal Sentinel-1 imagery.

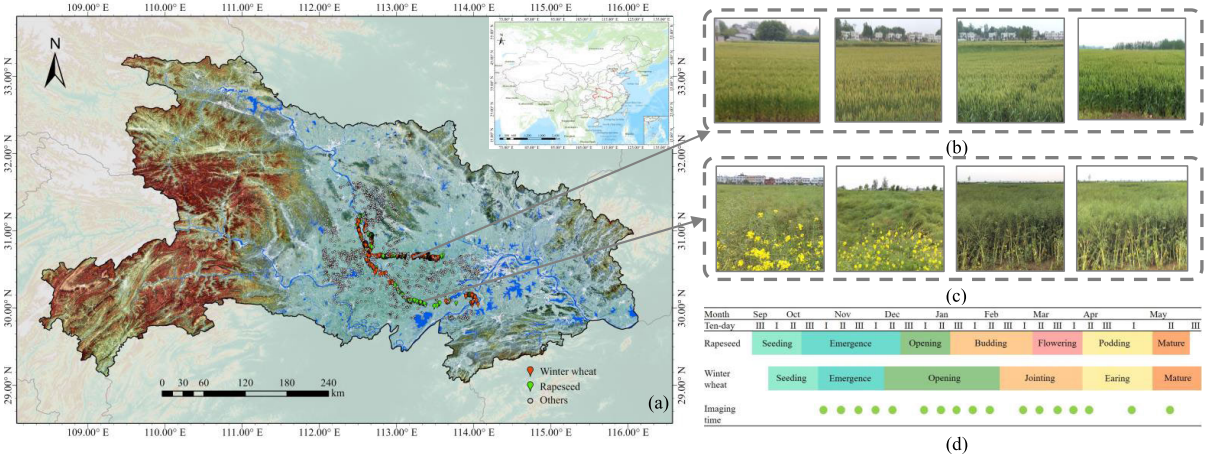


Fig. 5. (a) Geographical location of the study area, Hubei province in China. Photography of in situ samples of (b) winter wheat and (c) winter rapeseed in April 2021. (d) Phenological calendar for winter wheat and rapeseed, and the imaging time line of SAR SITS data.

B. Implement Details and Model Evaluation

The comparison methods include one ML method, i.e., RF [44], and four improved DL methods for crop mapping, i.e., multilayer LSTM [45], a domain-adapted RNN (DARNN) [46], DeepCropMapping (DCM) [31], and the transformer model [47]. RF, LSTM, and DARNN view each time step equally without the feature of key date extraction. DCM and transformer [24] extract key features from the whole crop growth stage, without considering the feature of the key phenological period.

The optimal hyperparameters for all the models were chosen in this study through grid search. For RF, the number of trees was 250. For DARNN, there were two 1-D convolutional layers of 4 and 8 filters with a kernel size of 1×3 and three RNN layers with 8, 16, and 32 hidden neurons. For LSTM and DCM, the number of LSTM layers is set to 2, and the number of hidden neurons in each LSTM unit is set to 256 and 512, respectively. For the transformer encoder, it consisted of two multihead self-attention layers with four heads, and 2048 hidden neurons in each attention module. For PhenoCropNet, the phenological period is set to 5 with 4, 3, 4, 3, and 3 sequences correspondingly, corresponding to the emergence, opening, budding, flowering, jointing, and maturity periods of the winter crop according to the winter crop phenological calendar in Hubei province. The number of BiGRUA modules is set to 5 and 1 for the two levels, respectively. There were 512 hidden neurons for each BiGRUA module in the hierarchical attention network, and also the classifier. The Adam optimizer was adopted for all DL methods, with the initial learning rate as 0.005, which was reduced to half when the total loss value stopped decreasing, taking 25 epochs as a step. L2 regularization was added, with λ being 10^{-4} for PhenoCropNet and 10^{-3} for the comparison methods. Each model was trained for 200 epochs, and the model with the best performance on the validation set was saved. The DL models were all run on a Linux platform (Ubuntu 7.2) with an Intel Xeon E5-2620 CPU, 64 GB of RAM, and four NVIDIA Tesla P100 graphics cards.

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The classification accuracy is evaluated on the total accuracy for all the class categories, represented as kappa and the individual accuracy for a certain class as $F1_i$, as shown in the following equations:

$$\text{kappa} = \frac{p_o - p_e}{1 - p_e} \tag{12}$$

$$p_o = \frac{1}{N}\sum_{i=1}^{C} n_i^{\text{corr}} \tag{13}$$

$$p_e = \frac{1}{N^2}\sum_{i=1}^{C} n_i^{\text{pred}} n_i^{\text{true}} \tag{14}$$

$$F1_i = 2 \cdot \frac{\text{PA}_i \cdot \text{UA}_i}{\text{PA}_i + \text{UA}_i} \tag{15}$$

$$\text{PA}_i = \frac{n_i^{\text{corr}}}{n_i^{\text{pred}}} \tag{16}$$

$$\text{UA}_i = \frac{n_i^{\text{corr}}}{n_i^{\text{true}}} \tag{17}$$

where p_o is the measure of the agreement between the observed and true labels, which is commonly referred to as the overall accuracy (OA), p_e is the measure of the expected agreement for a certain classifier, N is the overall number of samples, and C is the number of class categories. n_i^{corr} is the number of correctly classified samples, and n_i^{pred} and n_i^{true} are, respectively, the number of predicted samples and true samples for the ith category. PA_i and UA_i are the producer's accuracy and the user's accuracy for the ith category.

C. Comparison With Other Methods

1) Quantitative Evaluation: Table II shows the comparison of crop classification results between PhenoCropNet and the other ML and DL methods. This first demonstrates the feasibility of using SAR SITS data for dry-land crop differentiation in cloudy areas. The average accuracy of the method without focusing on the features of key phenological date (RF, LSTM, and DARNN) is around 0.9650 for winter wheat and 0.8872 for rapeseed. The average accuracy of the methods focusing on features of key phenological date while ignoring the features of key phenological period (DCM and transformer) is around 0.9651 for winter wheat and 0.8966 for rapeseed. This suggests the feature extraction of key phenological dates promotes the crop classification, especially for rapeseed. PhenoCropNet achieves the highest kappa, OA, and $F1$-score for the target crop, with the $F1$-score for winter wheat being above 0.97 and that for winter rapeseed being above 0.91, suggesting that the features of key phenological periods can further improve the crop classification of SAR SITS, and incorporating the phenological information of the crop growth stages and the hierarchical attention structure is stimulating to excavate phenological features of time-series SAR for crop classification.

2) Qualitative Evaluation: The advantage of temporal SAR crop mapping is that its active imaging mode enables it to obtain complete observation during the crop growth stage at a large scale, making seamless crop mapping available. Fig. 6(a) shows the winter crop mapping result for Hubei province based on PhenoCropNet using temporal SAR imagery. It is evident from the mapping result that the main producing area for winter wheat is intensively distributed in the center of the province, spanning north to south, and the south of Huangmei county. Compared with winter wheat, winter rapeseed is sparsely distributed in the province, presenting a distribution from east to west.

TABLE II
CLASSIFICATION ACCURACY COMPARISON RESULT OF THE PIXEL-WISE DL METHODS USING RAW SAR SITS DATA

Method	Winter wheat F1-score	Winter rapeseed F1-score	OA	Kappa
RF	0.9674	0.9123	0.9481	0.9294
LSTM	0.9668	0.9016	0.9399	0.9182
DARNN	0.9610	0.8478	0.9440	0.9236
DCM	0.9692	0.9068	0.9470	0.9278
Transformer	0.9609	0.8863	0.9454	0.9258
PhenoCropNet	**0.9708**	**0.9181**	**0.9519**	**0.9348**

Fig. 6(b)–(d) shows zoomed-in view crop mapping results in local regions of the main producing areas for the two winter crops of different methods. Fig. 6(b)–(d) shows that in the mixed cropping area, the fragmented parcel of winter crop can scarcely be identified by RF, which is obvious in Fig. 6(b), with rather coarse edge. While the edge of the parcel is rather smooth for DARNN, as shown in Fig. 6(b). For LSTM, DCM, and transformer which only use the time-series SITS in a global perspective, without considering the hierarchical relationship between different growth period, there are severe isolated misclassified pixels in the parcels. In the last row in Fig. 6(b) and (c), the edge of fragmented parcels can be maintained and the misclassification can be reduced for PhenoCropNet.

The features extracted by the DL models before the classifier and the original temporal VV and VH polarimetric signal through t-distributed stochastic neighbor embedding (t-SNE) transformation are visualized by scatter diagram in Fig. 7 to evaluate the efficiency of the feature extraction qualitatively. Each colored dot represents a sample of a certain category. For an effective classifier, the dots of the same color should be close, while the dots of different colors should be distant. Compared with the other DL methods, PhenoCropNet can extract more efficient features, which is evident in the more obvious boundaries between class categories and the fewer dots outside the clusters, especially for the two crop types. This further proves the effectiveness of the hierarchical attention module in PhenoCropNet for crop classification.

D. Visualization Analysis of PhenoCropNet

In this study, to measure the separability between two class categories before classification, the Jeffries–Matusita (JM) distance is used to calculate the separability between winter wheat and rapeseed, taking the SAR VV and VH signals at different single dates and in the different growth periods as the input to establish the key dates and periods. The closer the value approaches 2, the higher the separability is. Fig. 8(a) and (b) indicates the changes of the separability

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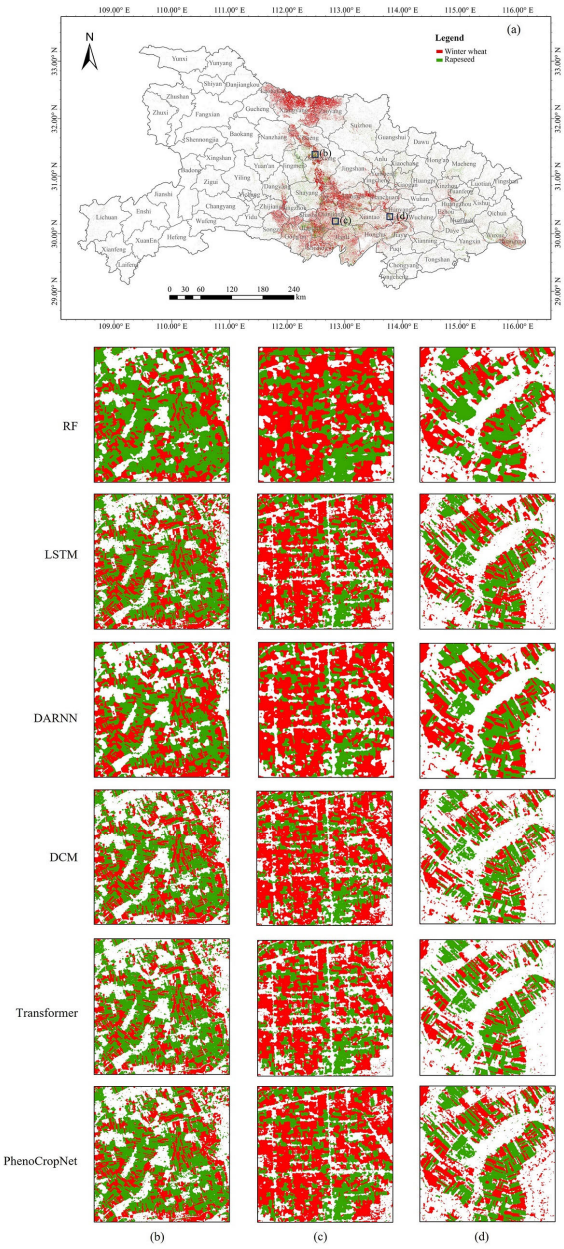


Fig. 6. (a) Winter crop mapping in Hubei province based on PhenoCropNet. (b)–(d) Zoomed-in view crop mapping result of RF, LSTM, DARNN, DCM, transformer, and PhenoCropNet.

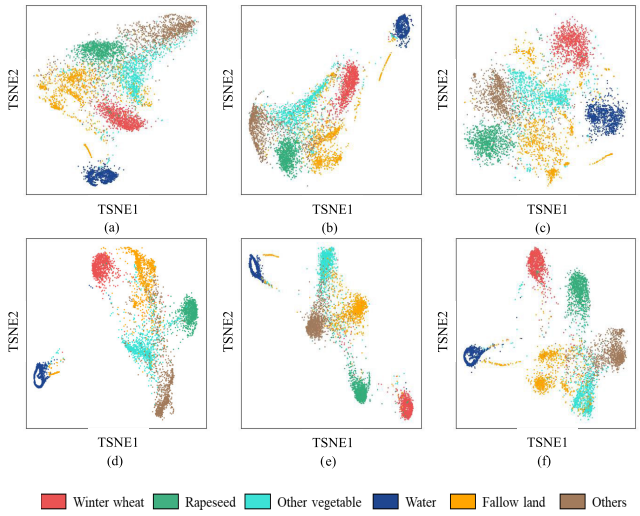


Fig. 7. Feature visualization through t-SNE of (a) raw temporal features of the VV and VH polarizations, and the features extracted by (b) LSTM, (c) DARNN, (d) DCM, (e) transformer model, and (f) PhenoCropNet, before input to the classifier.

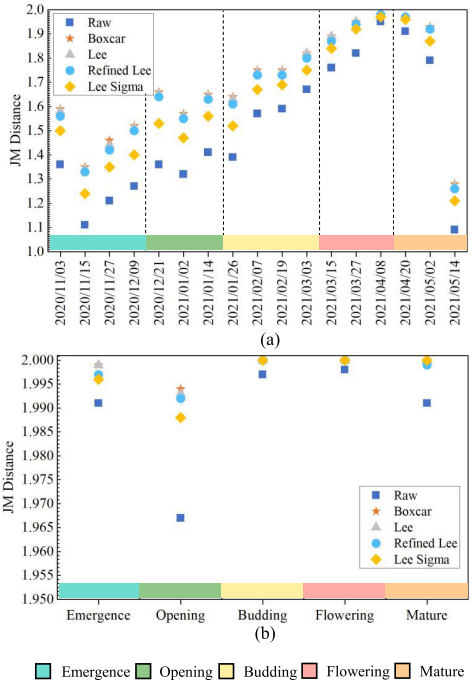


Fig. 8. JM distance between winter wheat and rapeseed of (a) each date and (b) growth period by taking time-series SAR of different temporal lengths as the input. Each section between the vertical dotted lines in (a) corresponds to a growth period in (b). The different colors in the bottom represent different crop growth periods in this study.

between different phenological dates and periods. A single date before the mature period is not enough to discriminate crops with JM distance under 2 in most cases, while key dates and temporal SAR in the growth period exert a positive effect on crop classification, with the obvious increase of JM distance. It shows the necessity to extract the effective classification features of important phenological dates and periods.

To further understanding the phenological pattern that PhenoCropNet has captured, the attention weights of different crop types at each date and in each growth period, as well as the impact of the growth period on crop classification are analyzed. The visualization of attention weights reveals which date and period PhenoCropNet has focused on when classifying the crop. Fig. 9(a) and (d) shows that for winter wheat, early growth periods, especially the flowering and mature period, are more important, with attention weights accounting for 90%. For key dates, during the emergence, opening, and budding period, the first date plays more important role with attention weights above 0.4, while for the flowering and mature period, all the dates in these periods account for equal importance. Fig. 9(b) and (e) shows that for winter rapeseed, growth periods of flowering and mature are more critical, with attention weights both above 0.4. And the attention weight of the dates during these crop growth periods are equally important, similar to winter wheat. This

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TABLE III
ABLATION EXPERIMENT RESULTS FOR THE PROPOSED PHENOCROPNET FRAMEWORK

	PAI	Hierarchical structure	Winter wheat F1-score	Winter rapeseed F1-score	OA	Kappa
A	×	×	0.9657	0.8888	0.9393	0.9173
B	√	×	0.9679	0.8930	0.9460	0.9266
C	√	√	**0.9708**	**0.9181**	**0.9519**	**0.9348**

suggests that the similarly equal attention weight of each date rather than the outstanding high attention weight of the single date might lead to the higher attention of certain growth period of winter wheat and wheat rapeseed. For other growth periods, including emergence, opening, and budding period, the last or penultimate date occupies the most attention. For other noncrop type, leisure land is taken as a representative example, as shown in Fig. 9(c) and (f). For leisure land, the attention weights of each date fluctuate less compared to crop types, remaining relatively stable. And the attention weights of the flowering and mature period account for a slightly higher percentage, around 0.06 higher than those of the emergence, opening, and budding period. The result is consistent to the previous studies [48, 40], which show that the jointing-heading and flowering stage, corresponding to the flowering period (mid-March to early April) in this study is key for winter wheat and winter rapeseed classification.

Moreover, to show the uncertainty that the growth periods bring to PhenoCropNet, in each experiment, the temporal sequence of different crop growth period was discarded, and the soft label value (probability) of each crop type was statistically represented as histogram, as shown in Fig. 10. The average soft label value is also plotted as the red dotted vertical line. For winter wheat, losing the growth period of flowering and mature has an adverse impact on the model, with the soft label value tend to be evenly distributed around 0.60–0.70. In contrast, losing the crop growth period of budding has the least impact, with most samples of the soft label value around 0.7–0.8. For winter rapeseed, losing the crop growth period of flowering and mature brings greater uncertainty to model classification, with the average probability lower by about 0.05 compared to losing the emergence, opening, and budding periods. In general, the impact of crop growth period corresponds to the importance of PhenoCropNet attention, which demonstrates that PhenoCropNet could concentrate on the discriminative periods for crop classification, and the key crop growth periods are significant for accurate crop classification.

IV. DISCUSSIONS

A. Effect of Various Modules

The ablation studies are designed to illustrate the effectiveness of the main modules in PhenoCropNet. The ablation experiments are planned in three groups—A, B, and C—as shown in Table III. Scheme A uses only one BiGRUA module as the network. Scheme B uses the PAI module and one BiGRUA module. Scheme C used the PAI module and hierarchical BiGRUA modules to extract the key features at different levels of phenological growth. The comparison between Schemes A and B suggests that the division on the basis of the crop phenological calendar is beneficial to the accuracy improvement by extracting the features at key dates in the different phenological periods. The division also helps to acquire the subtle differences for crops with similar phenological characteristics. The comparison between Schemes B and C proves the effectiveness of the hierarchical BiGRUA modules as it helps to fully explore the effect of the key period information.

B. Ability of Early Crop Mapping

The all-weather imaging ability of SAR enables it to acquire temporally consecutive backscattering of the ground, providing the chance for early crop identification without the disturbance of cloud. In the experiment in early crop identification, one SAR imagery at each date was added at a time in order until all the dates were used. Fig. 9 shows the addition of the temporal information has a positive influence on the crop identification, especially for rapeseed, which is evident in the improvement of ∼0.3 for winter wheat and ∼0.8 for rapeseed. The increase in the $F1$-score is prominently promoted with the addition of the new phenological periods, given the steeper slope from 2020/12/9 to 2020/12/21, 2021/1/26 to 2021/2/19, and 2021/3/15 to 2021/3/27, each of which is contained in a new growth period. In Fig. 11, the gray dotted line is the basic standard $F1$-score accuracy for early crop identification. For winter wheat, it can be identified as early as November of the previous year when the winter wheat is in the tilling stage, satisfying the need for yearly statistical surveying of the crop planting area before the end of the year, and accurately identified in February of the next year. For rapeseed, it can be identified in early February, when the rapeseed is in the flower bud development stage, which is earlier than the identification time for optical imagery, which requires imagery in the rapeseed flowering time. Similarly, rapeseed can be accurately identified up until late March, with the addition of more auxiliary temporal information. In general, the application of PhenoCropNet to SITS of SAR data is a feasible approach for early-season crop mapping.

C. More Applications in USA

To illustrate the generalization of PhenoCropNet, we have applied our proposed method in Iowa, USA, located in the Corn and Soybean Belt, for more crop type mapping. Temporal imagery from June 1st to September 30 in 2020 was used. The training, validation, and test samples were randomly selected from the Cropland Data Layer (CDL, https://www.nass.usda.gov/Research_and_Science/Cropland/sarsfaqs2.php). Considering the main crop types in Iowa are corn and soybean, the class categories are simplified by merging the similar land-cover types, finally retaining the following six categories, including corn, soybean, other vegetable, fallow land, water, impervious, and others. The $F1$-score of corn and soybean are 0.7012 and 0.6981, respectively, taking CDL as ground truth. The zoomed-in

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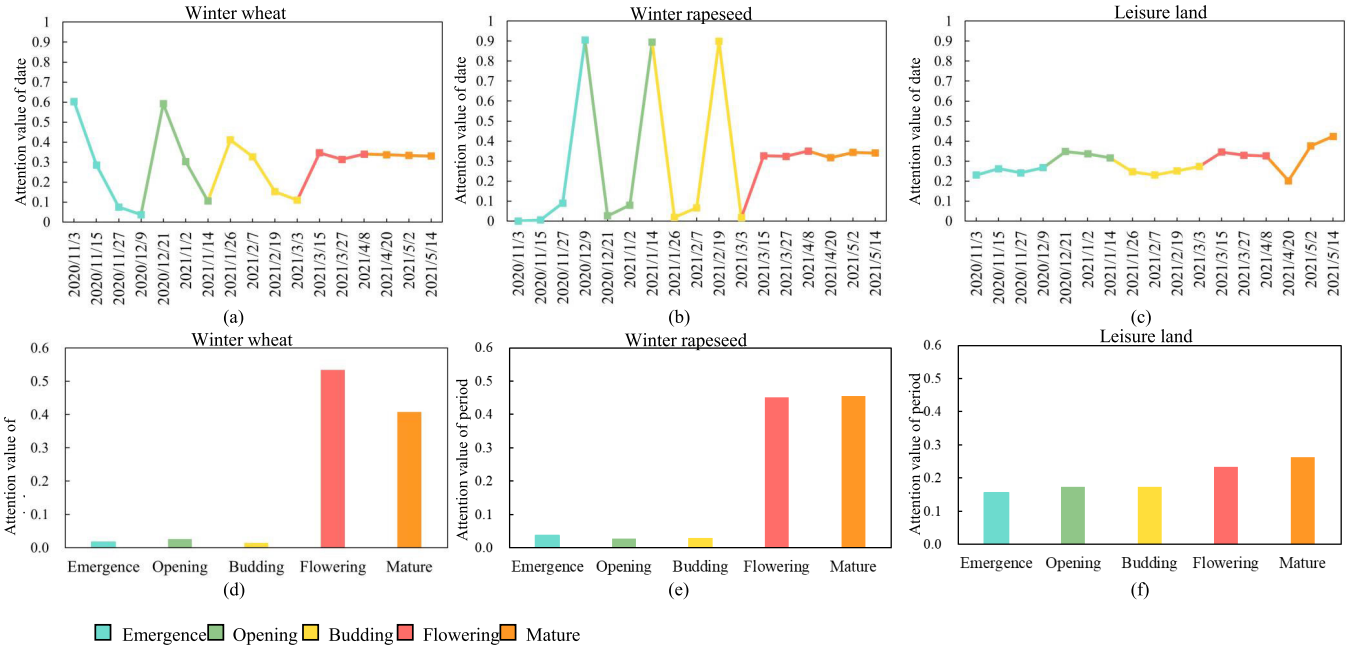


Fig. 9. Visualization of attention weights of key growth date of (a) winter wheat, (b) winter rapeseed, and (c) leisure land. Visualization of attention weights of key growth period of (d) winter wheat, (e) winter rapeseed, and (f) leisure land. Different colors of the line in (a)–(c) corresponds to the different crop growth period in (d)–(f).

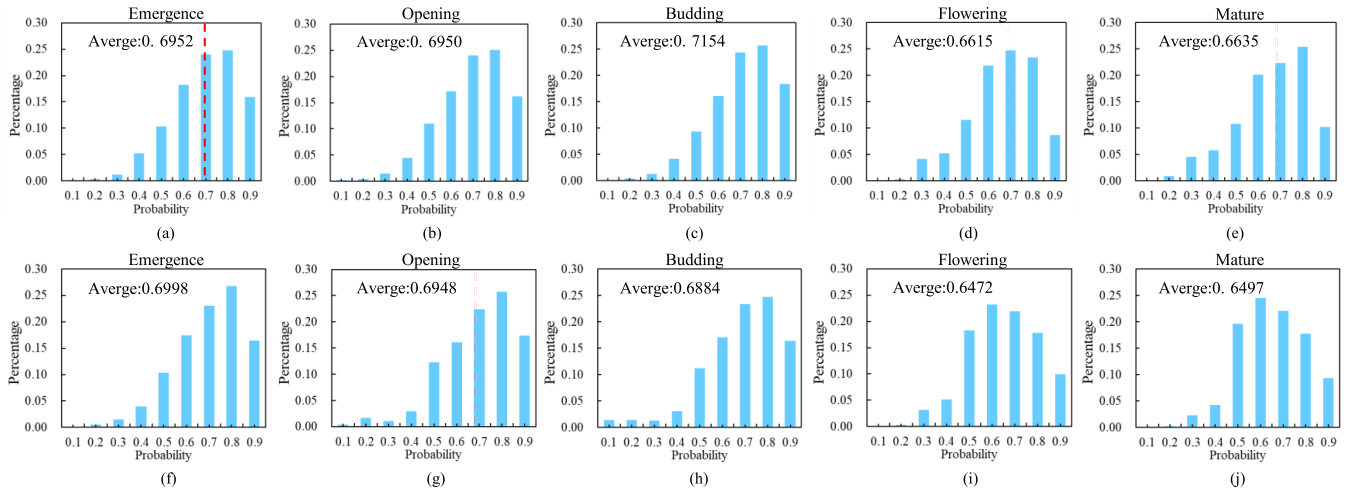


Fig. 10. Histogram of PhenoCropNet probability of (a)–(e) winter wheat and (f)–(j) winter rapeseed by discarding crop growth periods. The red dotted vertical line represents the average probability and the lower average probability indicates more importance of the corresponding growth period.

view and zoomed-out view crop mapping results are shown in Fig. 12. In general, the corn and soybean crop mapping result in Iowa keeps high consistency to CDL, especially in the north and south areas in Iowa, Fig. 12(b) and (d) shows. Although there are isolated misclassifications in the parcel in the middle area of Iowa, the general distribution of crop is correct compared with CDL. The crop mapping result in Iowa shows the great performance of PhenoCropNet to make full use of complete time-series SAR imagery for large-scale crop mapping and generalization performance of the model to be applied in different agricultural area for various crop type mapping.

D. Limitation and Prospect

The research objective of this study is how to make better use of phenological information of time-series SAR for crop classification in the absence of optical data for cloudy and rainy area, despite of the high accuracy it achieved, there are still space to improve in the future, coming from more phenological information addition and speckle reduction of SAR imagery. In the aspect of phenological information fusion, adding the information of crop growth period to time-series data and model improvement as this study proposes is a solution, while more phenological information related to crop growth is supposed to be added, such as growing degree days (GDDs), vegetable index derived from SAR. In the aspect of SAR speckle, although preprocess could reduce SAR speckle, there are isolated misclassified pixels. Object-oriented DL technology is supposed to be applied to reduce the misclassification pixels of the result on SAR imagery. Moreover, fusing optical imagery at key date also provides the solution to solve the problem.

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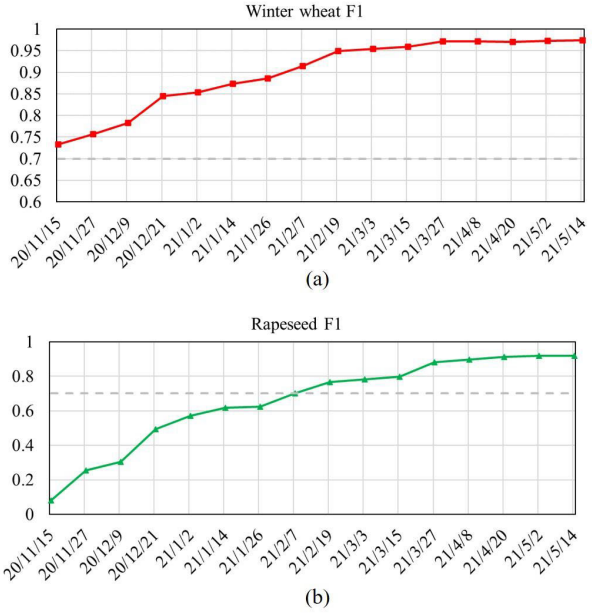


Fig. 11. $F1$-score of early crop identification accuracy for (a) winter wheat and (b) winter rapeseed using PhenoCropNet by taking SAR SITS as the input in an exhaustive way.

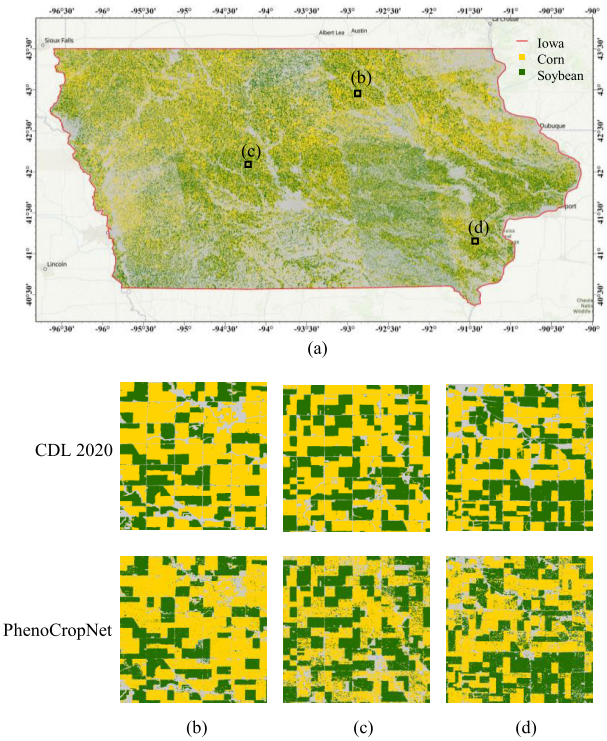


Fig. 12. Application of PhenoCropNet for crop mapping in Iowa, USA, in 2020. (a) Zoomed-out view crop mapping in Iowa. (b)–(d) Zoomed-in view crop mapping in local region.

V. Conclusion

In this article, a phenology-aware-based crop mapping network named PhenoCropNet has been proposed for large-scale crop mapping at cloudy areas by exploiting time-series SAR data. PhenoCropNet groups the time-series backscattering signal according to the crop phenological calendar, and introduces a hierarchical phenology self-attention network structure to extract the important features in two levels of the key date and key phenological periods to discriminate crop types with similar phenological characteristics. Differing from other DL models, PhenoCropNet considers both the phenological calendar information and prior knowledge about the important role of key dates and key periods for crop classification, obtaining the highest accuracy of winter crop classification compared with other ML methods. The visualization shows that PhenoCropNet can concentrate on important dates and periods by automatically adjusting the weight higher. Moreover, the visualization demonstrates that the key date and key growth periods for winter crop mapping are the date of obvious morphology change and the period of late maturity with stable conditions.

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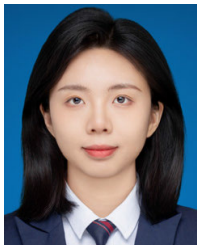
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Lei Lei received the B.S. degree from Southwest Jiaotong University, Chengdu, China, in 2020. She is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China.

Her research interests include deep learning and crop mapping.



Xinyu Wang (Member, IEEE) received the B.S. degree in photogrammetry and remote sensing from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2014, and the Ph.D. degree in communication and information systems from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, in 2019.

He is currently an Associate Research Fellow with the School of Remote Sensing and Information Engineering, Wuhan University. His research interests range from agriculture remote sensing, hyperspectral unmixing, and target detection.

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Xin Hu (Member, IEEE) received the B.S. degree in surveying and mapping engineering from Northeastern University, Shenyang, China, in 2017, and the Ph.D. degree in photogrammetry and remote sensing with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China, in 2022.

He is now a Senior Engineer with Guangzhou Urban Planning and Design Survey Research Institute, Guangzhou, China. His major research interests include agricultural remote sensing and hyperspectral image classification.



Liangpei Zhang (Fellow, IEEE) received the B.S. degree in physics from Hunan Normal University, Changsha, China, in 1982, the M.S. degree in optics from Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, in 1988, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1998.

He is a "Chang-Jiang Scholar" Chair Professor appointed by the Ministry of Education of China with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He was a Principal Scientist for the China State Key Basic Research Project (2011–2016) appointed by the Ministry of National Science and Technology of China to lead the remote sensing program in China. He has authored over 700 research articles and five books. He is an Institute for Scientific Information (ISI) Highly Cited Author. He currently holds 30 patents. His research interests include hyperspectral remote sensing, high-resolution remote sensing, image processing, and artificial intelligence.

Dr. Zhang is a fellow of the Institution of Engineering and Technology (IET). He received the 2010 Best Paper Boeing Award, the 2013 Best Paper ERDAS Award from the American Society of Photogrammetry and Remote Sensing (ASPRS), and the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). His research teams won the top three prizes of the IEEE GRSS 2014 Data Fusion Contest, and his students have been selected as the winners or finalists of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS) Student Paper Contest in recent years. He is the Founding Chair of the IEEE Geoscience and Remote Sensing Society (GRSS) Wuhan Chapter. He served as an Associate Editor or Editor for more than ten international journals. He is currently serving as an Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Yanfei Zhong (Senior Member, IEEE) received the B.S. degree in information engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

Since 2010, he has been a Full Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He has organized the Intelligent Data Extraction, Analysis and Applications of Remote Sensing (RSIDEA) Research Group. He has authored over 100 research articles in international journals, such as *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry*, *Remote Sensing*, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His research interests include hyperspectral remote sensing information processing, high-resolution remote sensing image understanding, and geoscience interpretation for multisource remote sensing data and applications.

Dr. Zhong is a fellow of the Institution of Engineering and Technology (IET). He received the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). He won the Second-Place Prize in the 2013 IEEE GRSS Data Fusion Contest and the Single-View Semantic 3-D Challenge of the 2019 IEEE GRSS Data Fusion Contest, respectively. He is currently serving as an Associate Editor for IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and the *International Journal of Remote Sensing*.

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